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|----------------|--|------|-----|
| 11 (a) | e.g. unreliable communication | (M1) | |
| | because ion layers vary in height / density | (A1) | |
| | e.g. cannot carry all information required | (M1) | |
| | bandwidth too narrow | (A1) | |
| | e.g. coverage limited | (M1) | |
| | reception poor in hilly areas | (A1) | |
| | <i>(any two sensible suggestions, M1 & A1 for each, max 4)</i> | | [4] |
|
(b) | signal must be amplified (greatly) before transmission back to Earth | B1 | |
| | <u>uplink</u> signal would be swamped by <u>downlink</u> signal | B1 | [2] |

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12 (a) (i) ratio / $\text{dB} = 10 \lg(P_1 / P_2)$

C1